

07 — Multi-Mode: Transit + Port

Proves · Modes never overlap in time, so each one runs its own cascade with the FULL budget.

Mechanics this scenario turns on

- Every active mode runs its **own** Level 1 — own demand, own combinations, own baseline, own optimum, own t/h. There is no single "tonnes per hour" for the vessel; the year is $\sum (\text{mode t/h} \times \text{mode hours})$.
- Cascade per row: H = what it wants · $I = \min(\text{remaining budget}, H)$ · $J = I \times \text{CoverageFactor}$ (covered) · $L = H - J$ (left to the gensets). Priority: DpReserve → DpDemand → Mission → Propulsion → Hotel. Invariant $\sum J + \sum L = \sum H$ — the sea sets the swing, the battery moves the split.
- With several modes the Power Demands tables gain one row per mode, and the header is an **hours-weighted average** (total energy ÷ total hours), not a sum.
- Level 2 and Level 3 are computed for **Transit only** (decision D4/Q5 — no workbook counterpart elsewhere). Other modes get an empty result, and the pinned-baseline radio does not reach them.

Panels described below · Battery Contribution — one cascade PER MODE, each with the FULL budget · Power Demands · Baseline & IL1 · Battery Benefit

Anything not described here behaves exactly as in [01-excel-baseline](#) — same plant, same hours; only what this scenario changes is worked through below.

Trust · verified against the reference workbook. These figures are proof.

Read after · scenario 01.

Test 01's vessel + Port 1 000 h / hotel 500. Battery 1260 for **both** Transit and Port.

Battery Contribution — one cascade PER MODE, each with the FULL budget

Modes never overlap in time (the ship is either at sea or at the quay), so each mode's cascade starts with the whole 1260 — the budget is not split.

- **Transit table:** identical to test 01 (573.15 → 200.6/372.5 · 76 → 3.8/72.2).
- **Port table:** a single row — no propulsion (moored), no mission. Hotel: $H = 500 \times 2\% = 10 \rightarrow I=10, J=0.5, L=9.5$; 1 250 kW of budget left unused.

Tiles are SUMS across modes: **SR = 444.7+9.5 = 454.2 · PS = 204.4+0.5 = 204.9.**

Power Demands

- Port row: propulsion 0, hotel 500+9.5 = 509.5 — carried by **SG 510 / AE 0**: the SG-forced rule makes the main engine run in port (~2 % load) to spin its shaft generator (logged observation #1; real vessels would run an AE).
- Energy: Transit 75 427 738 + Port 509.5×1 000 = 509 500 → header ME = 81 622 000/... actually (75 427 738+509 500)/6 000 = **12 656** · AE 3 111 000/6 000 = **518**.

Baseline & IL1

Transit part identical to 01; Port adds ~equal amounts to both sides: Baseline **13 396.6** · IL1 **13 375.4** · savings **21.2 t** (unchanged — port has nothing to optimize in this plant).

Battery Benefit

Per-mode dual scenarios, summed: Transit 173.66 + Port $\approx$ 0.09 = **173.74 t/yr = \$135 520**. The port contribution is ~\$68/year — the model honestly says a battery bought for a 500-kW port hotel is pointless.

Takeaway: per-mode full budget, per-mode cascade shape (port = one row), tiles and benefit are sums — and the SG-forced quirk becomes directly visible in the Port row.

KSailCalc manual test scenarios — calculation walkthrough · regenerated 2026-08-07 · numbers verified against commit 559aef2 and unchanged by the backend/client refactors (18 golden snapshots byte-identical)